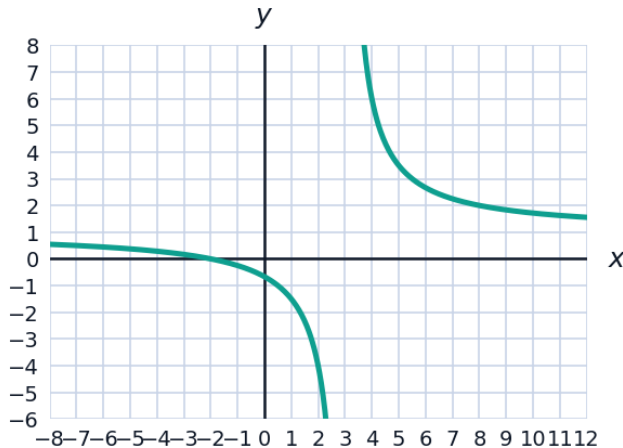


Graphing Rational Functions



Reading graphs

- 1 The graph of $f(x) = \frac{x+2}{x-3}$ is shown. From the graph and the formula, state the asymptotes and intercepts.



- 2 The function $g(x) = \frac{1}{x-2} + 3$ is a transformation of $y = \frac{1}{x}$.
- Describe the transformation.
 - State the asymptotes of g .
 - State the domain and range.

Sign analysis

- 3 Let $f(x) = \frac{x-1}{(x+2)(x-4)}$. Using a sign chart with critical values $x = -2, 1, 4$, determine the intervals on which $f(x) > 0$.
- 4 Solve the rational inequality $\frac{x+3}{x-2} \leq 0$.
- 5 Sketch-planning: for $h(x) = \frac{2x}{x^2-1}$, state the asymptotes and the symmetry of the graph.

Sketching a complete graph

- 6 For $f(x) = \frac{x^2-1}{x^2-4}$, find all intercepts and asymptotes, then describe the graph's behaviour near each vertical asymptote.
- 7 Sketch-planning: the function $g(x) = \frac{3}{x-1} - 2$ has a vertical asymptote at $x = 1$ and horizontal asymptote $y = -2$. Find its x - and y -intercepts.

More rational inequalities

8 Solve $\frac{(x-2)(x+1)}{x-4} \geq 0$.

9 Solve $\frac{2}{x+3} > 1$.